

Subodh Sonar

Scientist | Process & Automation Engineer | Materials Researcher

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PROFILE

Experienced process engineer and materials scientist with 5+ years in biotech, specializing in sustaining, process scale-up, and Hamilton liquid handling. Proven track record in significant cost savings through designing data-driven solutions for operational efficiency including automation and product shelf life extension. Skilled in Six Sigma Manufacturing, pilot-scale operations, process transfers, and troubleshooting. Adept at developing SOPs, drafting batch records, and managing engineering changes. Strong collaborator and mentor. Enjoys complex challenges and delivers impactful results while optimizing resources. Highly motivated, curious and adaptable.

WORK EXPERIENCE

Senior Research Associate I/ II, 10x Genomics | Pleasanton, CA | July 2021 - Present

- Developed a cleaning and drying-in-place protocol for a custom dispensing robot, saving 180 labor hours and \$200K annually. Replaced complex assays with a total organic carbon (TOC) based QC method, reducing testing costs by \$3,000 per run.
- Reduced scrap costs by \$1.4M by mitigating process variation in multi-day oligo-ligation processes for Chromium Single Cell consumables. Identified key failure modes linked to chemical interactions in polymer and oligo chains on microfluidic chip performance.
- Extended shelf life of Chromium Single Cell gel bead reagents from 2 to 4 years by validating real-time data against original QC results, saving ~\$4M in inventory costs.
- Alleviated gel bead tethering impacts on Chromium chips. Helped implement novel reagent composition, improving temperature stability and reducing temperature related failure rates from 4% to <1%.
- Led cold-chain shipping studies and developed an optimized packing ratio to maintain < -40°C for 5-day transit, eliminating temperature excursions within re-icing schedules to ensure gel bead product integrity.
- Developed a non-destructive, high-throughput computer vision QC to replace cytometry-based method, boosting throughput. Enabled a discovery that reduced required gel bead concentration for optimal chip performance.
- Created a Google Sheets dashboard with Python/Streamlit backend to visualize oligo yield data, enabling real-time manufacturing insights via dynamic heatmaps and plots.
- Supported scale-up and tech transfer for Chromium and Xenium consumables, authoring batch records, SOPs, and QCMs; used DoEs, SPC, FMEAs, and Gage R&Rs to validate processes.
- Optimized Xenium dispensing, sealing, and shipping workflows, cutting customer complaints by 80% in two quarters and improving dry-ice packing efficiency.
- Authored FAT/SAT protocols for new sites to ensure consistent process validation, timelines, and quality checks across manufacturing locations.

Scientist I/II, Becton Dickinson Biosciences | La Jolla, CA | August 2019 - June 2021

- Led experimentation for a raw material blending strategy for fluorescent dye synthesis, resulting in patented IP (USP 20220340813A1). Developed tunable dyes with enhanced optical properties for BD FACS reagents.
- Conducted multiple DoEs to optimize chain-growth polymerization, controlling reaction pathways and kinetics using SPC and GPC. Achieved repeatability and reproducibility across scales, reducing aggregation and background issues in Brilliant Violet dye reagent lots.
- Performed an MSA and Gage R&R on NanoDrop Spectrophotometer, revealing a 30% variance in absorbance causing significant process instability. Proposed adjustments for improved calibration and enhanced quality.
- Eliminated a lyophilization step in dye production, saving \$12,000 per lot in labor and product loss. Used Python for data scraping and Minitab to visualize process trends.
- Identified and qualified alternative polymer purification consumable suppliers to mitigate risk from product discontinuation, ensuring continuity of operations and supply chain resilience.

AUTOMATION EXPERIENCE

10x Genomics | Hamilton STAR, STARlet, Vantage

- Scripted methods for scaled-down oligo-ligation chemistry with modular functions for R&D. Implemented custom tip adapter and pickup pattern to achieve barcode diversity. Optimized liquid class for viscous gel-bead solutions in aqueous buffers, saving 8 hours per iteration and reducing ergonomic stress on operators.
- Assisted in automating manual dispensing processes, creating SOPs, conducting FMEA, and ensuring QC sampling, saving up to 48 hours of labor per quarter. Designed and executed a cross-contamination study for automated dispensing in next-gen Xenium plates, optimizing liquid classes for oligonucleotide solutions. Used tracer dye and fluorescence to ensure <0.1% cross-contamination during 10 consecutive dispenses.

BD Biosciences | ML apps, Opentrons

- Developed an ML app for predicting functional group exchange reactions, achieving 4% mean squared error. Modeled using SVM, XgBoost and RF Regressors. Deployed using Gradio, saving 16 hours of R&D per screening run and reducing QC testing load.

RESEARCH EXPERIENCE

Research Internship, BD Biosciences | La Jolla, CA | June 2018 - September 2018

- Performed GPC-based polymer analysis to optimize fluorescent polymer synthesis, developing calibration curves for polymerization and functionalization, for enhanced fluorophore conjugation chemistry and yield.

Graduate Researcher, University of California San Diego | January 2018 - June 2019

- Conducted interdisciplinary research in advanced materials as part of a MURI-funded project. Focused on polydopamine nanoparticle synthesis, mixed-matrix membranes, and bio-inspired chemical systems, achieving high absorptive thin films for solar energy applications.
- ECE136L: Fabricated diodes, LEDs, MOS capacitors, and FETs via photolithography and thin-film deposition for semiconductor applications, with analysis using voltametry and AFM.
- ECE202: Conducted characterization of implantable sensing electrodes, designed EEG/EMG arrays using CAD - printed using laser micro machining, and performed neural data analysis.

Undergraduate Researcher, Institute of Chemical Technology | October 2016 - September 2017

- Designed, synthesized and analyzed Cerium and Zirconium-based MOFs (UiO-66/808) and APTES-functionalized MCM-41 catalysts, achieving >90% conversions in 30+ catalytic reactions such as transamination, formylation and CO₂ cycloaddition.

Summer Researcher, Department of Chemical Engineering | IIT Bombay | 2015

- Assessed small molecule recovery from polymeric by-products (Asphaltenes) of refined crude oil, estimating molecular weight through osmolality measurements and HPTLC analysis. Classified macromolecule constituents and identified separable organic compounds using polarity-based solvent mixtures.

EDUCATION

University of California, San Diego

M.S. in Materials Science and Engineering, June 2019

Institute of Chemical Technology (UDCT), Mumbai

B.S. in Chemical Engineering, June 2017

HIGHLIGHTS

United States Patent 20220340813A1 Water-Soluble Fluorescent Polymeric Dyes BD | Oct 2022

Certified **Venus Software I** for Hamilton Liquid Automation Systems 10x Genomics | April 2024

Certified in **Six Sigma Manufacturing** Fundamentals Design BD | November 2019

Winning author, "*Proposal to study MOFs using the Synchrotron beam line (26-ID-C) at ANL*" UCSD | Spring 2018

CORE COMPETENCIES

Analytical Tools: HPLC, GC-MS, GPC, HPTLC, XRD (Bruker), SEM-EDS, FTIR, BET, XPS, AFM, ¹H-NMR, UV/Vis

Automation & Scripting: Hamilton Venus 6, OpenCV, C, C++, Docker, Gradio, Labview, Opentrons, R-studio

Software & Data Analysis: JMP, Minitab, Tableau, Python (Pandas, Scikit-learn, Streamlit), CAD, Solidworks

Process Development: DOEs, SPC, Scale-up, Gage R&R, Cleaning validations, NPI tech-transfer, FMEA, RCAs

Regulatory & Quality: GxP, Equipment validation, SOP development, QMS, Engineering change management